



Kartikey Rai

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EDUCATION

Degree/Certificate	Institute/Board	Year
B.Tech Cse	Indian Institute of Information Technology, Una	2022-Present
Senior Secondary	Board of High School and Secondary Education , U.P.	2020

EXPERIENCE

TechVizor (SDE Intern)

dec 2025 - present

Python, FastAPI, WebSockets, Groq LLM, Prompt Engineering, Faster-Whisper STT, Edge-TTS, Js

on-site

- Developed **AI-powered voice and text chatbot** for employee management using **FastAPI, Groq LLM, Faster-Whisper STT**, and **Edge-TTS** with **real-time streaming** capabilities.
- Implemented **shared session management system** maintaining conversation context across voice and text modes, enabling seamless interaction without data loss.
- Designed **intelligent timeout management** with adaptive timers that pause during processing and reset on user input for natural conversational flow.
- Integrated **employee profile data** for personalized, context-aware responses achieving **sub-2-second latency** for HR self-service operations.

SNACKBAE (AI Intern)

March 2025 - June 2025

Python, LangChain, Groq LLM, HuggingFace Embeddings, FAISS, NLP, RAG, Prompt Engineering

Remote

- * Developed a **Generative AI-powered chatbot tutor** using LLMs and RAG pipelines, enabling students to access **24/7 context-aware support** and reducing instructor queries by **40%**.
- * Automated **quiz and flashcard generation** with NLP, cutting manual content creation time by **60%** and enhancing learner engagement through interactive assessments.
- * Implemented **prompt engineering and response optimization**, achieving a **30% improvement in answer accuracy** and reducing model hallucinations significantly.
- * Built and integrated the solution using **Python and LangChain**, supporting **500+ active learners** with a scalable and responsive e-learning platform.

PROJECTS

Multi-Source AI Search Assistant 🗕🔗*

Generative AI, Agentic AI, Python, Streamlit, LangChain, Groq API, Arxiv API, Wikipedia API, DuckDuckGo Search

- Developed an AI-powered web search assistant using LangChain, enabling users to query Arxiv, Wikipedia, and the web in real-time.
- Integrated a Groq LLM for reasoning and generating concise, context-aware answers from multiple sources.
- Implemented Streamlit frontend with streaming responses and interactive chat for improved user engagement.
- Handled multi-source data retrieval and error parsing to ensure accurate and reliable information delivery.

Conversational PDF Chatbot (RAG System) 🗕🔗*

Python, Streamlit, LangChain, Groq API, HuggingFace Embeddings, FAISS, PyPDFLoader

- Developed an interactive web application enabling users to upload PDFs and chat with the content using a conversational AI model.
- Integrated Retrieval-Augmented Generation (RAG) to provide context-aware answers, reformulating user queries for improved understanding.
- Integrated multi-turn chat history and feedback logging with latency tracking for quality monitoring.
- Optimized AI responses using custom prompts to provide concise, accurate answers in under three sentences.

SKILLS

- **Programming Languages:** C, C++, Python
- **Generative AI :** FastAPI, LLM, Prompt Engineering, LangChain, Groq API, HuggingFace Embeddings, ChromaDB, FAISS, Pinecone, Transformers
- **Deep Learning & NLP:** Neural Networks, CNN, RNN, LSTM, Attention Mechanism, Transformers, Transfer Learning
- **Core CS Skills:** Data Structures & Algorithms (C++), Object-Oriented Programming, Database Management Systems
- **Tools & Platforms:** Git, GitHub, Streamlit, docker
- **Other Skills:** Problem solving, communication

ACHIEVEMENTS

Deep Learning Expertise: Implemented advanced models including CNN, RNN, LSTM, and Transformers for Natural Language Processing and Generative AI tasks.

DSA in C++: Solved 600+ problems in Data Structures & Algorithms, strengthening algorithmic thinking and problem-solving skills.